Chapter 4. Randomized Blocks, Latin Squares, and Related Designs

Contents

1 Introduction and the Blocking Principle

- Topics: blocking and nuisance factors; the randomized complete block design (RCBD); extension of ANOVA to the RCBD; other blocking scenarios — Latin squares, Graeco-Latin squares, balanced incomplete block designs (BIBD).
- **Blocking** is a technique for dealing with **nuisance factors**: factors that affect the response but are of no interest to the experimenter; the variability they transmit must be minimized.
- Typical nuisance factors: batches of raw material, operators, pieces of test equipment, time (shifts, days), different experimental units. Failure to block is a common flaw in designing an experiment.

Nuisance factor is ...	Strategy
known and controllable	**blocking**
known but uncontrollable (measurable)	**analysis of covariance** (Ch. 11)
unknown and uncontrollable (lurking)	**randomization** (hope it balances out)

2 Hardness Testing Example

- Do four different tips produce different mean hardness readings on a Rockwell tester?
- Tips are applied to test coupons. A CRD would assign tips to coupons at random — but coupons differ in hardness, a source of nuisance variability; the engineer may also want to test tips across coupons of various hardness.
- **RCBD**: assign *all four tips to each coupon.* Each coupon is a **block** — a more homogeneous experimental unit. Variability *between* blocks can be large; variability *within* a block should be small. A complete replicate of the experiment is run in each block; a block is a restriction on randomization (runs are randomized *within* a block).

	Test coupon (block)				
Tip	1	2	3	4	$y_{i\cdot}$
1	9.3	9.4	9.6	10.0	38.3
2	9.4	9.3	9.8	9.9	38.4
3	9.2	9.4	9.5	9.7	37.8
4	9.7	9.6	10.0	10.2	39.5
$y_{\cdot j}$	37.6	37.7	38.9	39.8	$y_{\cdot\cdot} = 154.0$

The data have a two-way structure; we test equality of treatment means after removing block variability.

3 The RCBD Model and ANOVA

3.1 Model

Definition 4.1: RCBD effects model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, \dots, a;\ j = 1, \dots, b,$$

$\sum_i \tau_i = 0$, $\sum_j \beta_j = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Means model: $y_{ij} = \mu_{ij} + \varepsilon_{ij}$, $\mu_{ij} = \mu + \tau_i + \beta_j$ (additive: no treatment×block interaction).

$H_0 : \mu_1 = \cdots = \mu_a$ $(\mu_i = \frac{1}{b}\sum_j \mu_{ij} = \mu + \tau_i)$ $\iff$ $H_0 : \tau_1 = \cdots = \tau_a = 0$.

3.2 Partition of the total sum of squares

Theorem 4.2

$$\sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = b\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + a\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$$

$$SS_T = SS_{\text{Treatments}} + SS_{\text{Blocks}} + SS_E,$$

$$ab - 1 = (a-1) + (b-1) + (a-1)(b-1).$$

Proof. Write $y_{ij} - \bar{y}_{\cdot\cdot} = A_i + B_j + C_{ij}$ with $A_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $B_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $C_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Squaring and summing gives the three squares plus three cross products, each of which vanishes:

$$\sum_i\sum_j A_i B_j = \Big(\sum_i A_i\Big)\Big(\sum_j B_j\Big) = 0 \cdot 0 = 0,$$

$$\sum_i\sum_j A_i C_{ij} = \sum_i A_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}) = \sum_i A_i (b\bar{y}_{i\cdot} - b\bar{y}_{i\cdot} - b\bar{y}_{\cdot\cdot} + b\bar{y}_{\cdot\cdot}) = 0,$$

$$\sum_i\sum_j B_j C_{ij} = \sum_j B_j \sum_i (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}) = \sum_j B_j (a\bar{y}_{\cdot j} - a\bar{y}_{\cdot\cdot} - a\bar{y}_{\cdot j} + a\bar{y}_{\cdot\cdot}) = 0.$$

Since A_i does not depend on j, $\sum_i \sum_j A_i^2 = b\sum_i A_i^2$; similarly $\sum\sum B_j^2 = a\sum_j B_j^2$. Degrees of freedom of SS_E: the ab residuals C_{ij} satisfy $\sum_i C_{ij} = 0$ (b constraints) and $\sum_j C_{ij} = 0$ (a constraints, one redundant), so df $= ab - (a + b - 1) = (a-1)(b-1)$. ∎

Source	SS	DF	MS	F_0
Treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Blocks	SS_{Blocks}	$b-1$	MS_{Blocks}	
Error	SS_E	$(a-1)(b-1)$	MS_E	
Total	SS_T	$N-1$		

Reject H_0 if $F_0 > F_{\alpha, a-1, (a-1)(b-1)}$.

Derivation: what about $H_0 : \beta_j = 0$?

Testing blocks is generally *not* of interest (blocks were used precisely because we believe they differ), and because of the randomization restriction the ratio MS_{Blocks}/MS_E is only an approximate indicator of block importance. If MS_{Blocks}/MS_E is large, blocking was worthwhile.

3.3 Expected mean squares (elementwise)

Theorem 4.3

$$\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \frac{b\sum_i \tau_i^2}{a-1}, \qquad \mathrm{E}(MS_{\text{Blocks}}) = \sigma^2 + \frac{a\sum_j \beta_j^2}{b-1}, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Treatments. $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\beta}_\cdot + \bar{\varepsilon}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and $\bar{y}_{\cdot\cdot} = \mu + \bar{\varepsilon}_{\cdot\cdot}$. Hence $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$ — the block effects cancel. Then

$$\begin{aligned}
\mathrm{E}(MS_{\text{Treatments}}) &= \frac{b}{a-1}\mathrm{E}\Big(\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2\Big) = \frac{b}{a-1}\mathrm{E}\Big(\sum_i \big(\tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})\big)^2\Big) \\
&= \frac{b}{a-1}\Big[\sum_i \tau_i^2 + \mathrm{E}\sum_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 + 2\underbrace{\mathrm{E}\sum_i \tau_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})}_{=0}\Big] \\
&= \frac{b}{a-1}\sum_i \tau_i^2 + \frac{b}{a-1}\cdot(a-1)\underbrace{\mathrm{E}\Big[\frac{\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}{a-1}\Big]}_{=\mathrm{Var}(\bar{\varepsilon}_{i\cdot}) = \sigma^2/b} = \sigma^2 + \frac{b}{a-1}\sum_{i=1}^{a}\tau_i^2.
\end{aligned}$$

By symmetry (swap the roles of i, a, τ and j, b, β), $\mathrm{E}(MS_{\text{Blocks}}) = \sigma^2 + \frac{a}{b-1}\sum_j \beta_j^2$. ■

Error. Substituting the model, μ, τ_i, β_j all cancel:

$$y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot j} + \bar{\varepsilon}_{\cdot\cdot} = (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) - (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}).$$

So

$$SS_E = \sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2 + \sum\sum(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2 - 2\sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}).$$

① For each i, $\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2/(b-1)$ is a sample variance of b iid errors:

$$\mathrm{E}\Big[\sum_i\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2\Big] = \sum_{i=1}^{a}\mathrm{E}\Big[\frac{\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{b-1}\Big](b-1) = a(b-1)\sigma^2.$$

② $\bar{\varepsilon}_{\cdot 1}, \ldots, \bar{\varepsilon}_{\cdot b}$ are iid with variance σ^2/a:

$$\mathrm{E}\Big[\sum_i\sum_j(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2\Big] = a\,\mathrm{E}\Big[\frac{\sum_j(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2}{b-1}\Big](b-1) = a\cdot\frac{\sigma^2}{a}(b-1) = (b-1)\sigma^2.$$

③ Expand the cross product into four pieces:

$$\sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}) = \underbrace{\sum\sum\varepsilon_{ij}\bar{\varepsilon}_{\cdot j}}_{(i)} - \underbrace{\sum\sum\varepsilon_{ij}\bar{\varepsilon}_{\cdot\cdot}}_{(ii)} - \underbrace{\sum\sum\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j}}_{(iii)} + \underbrace{\sum\sum\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot\cdot}}_{(iv)}.$$

(i) $\sum_j \bar{\varepsilon}_{\cdot j}\sum_i \varepsilon_{ij} = a\sum_j \bar{\varepsilon}_{\cdot j}^2$, so $\mathrm{E}(i) = a\cdot b\cdot\frac{\sigma^2}{a} = b\sigma^2$.

(ii) $\bar{\varepsilon}_{\cdot\cdot}\sum\sum\varepsilon_{ij} = ab\,\bar{\varepsilon}_{\cdot\cdot}^2$, so $\mathrm{E}(ii) = ab\cdot\frac{\sigma^2}{ab} = \sigma^2$.

(iii) $\sum_i\sum_j \bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j} = (\sum_i \bar{\varepsilon}_{i\cdot})(\sum_j \bar{\varepsilon}_{\cdot j}) = a\bar{\varepsilon}_{\cdot\cdot}\cdot b\bar{\varepsilon}_{\cdot\cdot}$, so $\mathrm{E}(iii) = ab\frac{\sigma^2}{ab} = \sigma^2$. (Directly: $\mathrm{E}(\bar{\varepsilon}_{1\cdot}\bar{\varepsilon}_{\cdot 1}) = \mathrm{E}\big[\frac{1}{b}(\varepsilon_{11}+\cdots+\varepsilon_{1b})\frac{1}{a}(\varepsilon_{11}+\cdots+\varepsilon_{a1})\big] = \frac{1}{ab}\mathrm{E}(\varepsilon_{11}^2) = \frac{\sigma^2}{ab}$, and there are ab such terms.)

(iv) Same as (ii) by symmetry: σ^2.

Thus $\mathrm{E}③ = b\sigma^2 - \sigma^2 - \sigma^2 + \sigma^2 = (b-1)\sigma^2$, and

$$\mathrm{E}(SS_E) = a(b-1)\sigma^2 + (b-1)\sigma^2 - 2(b-1)\sigma^2 = (a-1)(b-1)\sigma^2 \;\Rightarrow\; \mathrm{E}(MS_E) = \sigma^2.$$

■

Remark 4.4: Matrix shortcut

With $\mathbf{H}$ the hat matrix of the additive model ($\mathrm{rank}\mathbf{X} = 1+(a-1)+(b-1) = a+b-1$), $SS_E = (\mathbf{y}-\hat{\mathbf{y}})^T(\mathbf{y}-\hat{\mathbf{y}}) = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$ and by Theorem 3.13

$$\mathrm{E}(SS_E) = \sigma^2\mathrm{tr}(\mathbf{I}-\mathbf{H}) + \underbrace{\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta}}_{=0} = \sigma^2(ab-(a+b-1)) = (a-1)(b-1)\sigma^2.$$

For $a=b=2$: $\mathbf{y} = (y_{11},y_{12},y_{21},y_{22})^T$, $\boldsymbol{\beta} = (\mu,\tau_1,\beta_1)^T$, $\mathbf{X} = \begin{bmatrix}1&1&1\\1&1&-1\\1&-1&1\\1&-1&-1\end{bmatrix}$; dropping the block column (if $\beta_1=0$) gives $\mathbf{X} = \begin{bmatrix}1&1\\1&1\\1&-1\\1&-1\end{bmatrix}$, $\boldsymbol{\beta} = (\mu,\tau_1)^T$.

3.4 Least squares estimation

Derivation

$S = \sum_i\sum_j(y_{ij}-\mu-\tau_i-\beta_j)^2$. Setting $\partial S/\partial\theta|_{\hat\theta} = 0$:

$$\frac{\partial S}{\partial\mu}: \; -2\sum_i\sum_j(y_{ij}-\hat\mu-\hat\tau_i-\hat\beta_j) = 0 \;\Rightarrow\; ab\hat\mu + b\sum\hat\tau_i + a\sum\hat\beta_j = y_{\cdot\cdot} \;\Rightarrow\; \hat\mu = \bar y_{\cdot\cdot},$$

$$\frac{\partial S}{\partial\beta_j}: \; -2\sum_{i=1}^{a}(y_{ij}-\hat\mu-\hat\tau_i-\hat\beta_j) = 0 \;\Rightarrow\; a\hat\mu + \textstyle\sum_i\hat\tau_i + a\hat\beta_j = y_{\cdot j} \;\Rightarrow\; \hat\beta_j = \bar y_{\cdot j} - \bar y_{\cdot\cdot},$$

$$\frac{\partial S}{\partial\tau_i}: \; -2\sum_{j=1}^{b}(y_{ij}-\hat\mu-\hat\tau_i-\hat\beta_j) = 0 \;\Rightarrow\; b\hat\mu + b\hat\tau_i + \textstyle\sum_j\hat\beta_j = y_{i\cdot} \;\Rightarrow\; \hat\tau_i = \bar y_{i\cdot} - \bar y_{\cdot\cdot},$$

using $\sum\hat\tau_i = \sum\hat\beta_j = 0$. Fitted values and residuals:

$$\hat y_{ij} = \hat\mu + \hat\tau_i + \hat\beta_j = \bar y_{i\cdot} + \bar y_{\cdot j} - \bar y_{\cdot\cdot}, \qquad e_{ij} = y_{ij} - \bar y_{i\cdot} - \bar y_{\cdot j} + \bar y_{\cdot\cdot}.$$

Note $E(\hat\tau_i) = \mathrm{E}(\bar y_{i\cdot}) - \mathrm{E}(\bar y_{\cdot\cdot}) = \frac{1}{b}(b\mu + b\tau_i + \sum_j\beta_j) - \frac{1}{ab}(ab\mu + b\sum\tau_i + a\sum\beta_j) = \tau_i$: in a *complete* block design every treatment sees every block, so the block effects average out. (Contrast with the BIBD in Section 6.)

3.5 Extra sum of squares derivation of the F-test

Derivation: extra sum of squares method

Full model (F): $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$; Reduced model (R) under $H_0 : \tau_i = 0\ \forall i$: $y_{ij} = \mu + \beta_j + \varepsilon_{ij}$.

$$SS_{E(F)} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2, \qquad SS_{E(R)} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot j})^2,$$
$$SS_H = SS_{E(R)} - SS_{E(F)}.$$

Theorem 4.5: $SS_H = SS_{\text{Treatments}}$

Proof. Write $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{\cdot j}) - (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})$. Then

$$\begin{aligned} SS_{E(F)} &= \sum\sum (y_{ij} - \bar{y}_{\cdot j})^2 + \sum\sum (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2\sum\sum (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(y_{ij} - \bar{y}_{\cdot j}) \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \underbrace{\frac{\sum_j (y_{ij} - \bar{y}_{\cdot j})}{b}}_{=\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}} \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{E(R)} - b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2. \end{aligned}$$

Hence $SS_H = SS_{E(R)} - SS_{E(F)} = b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{\text{Treatments}}$. ■

Matrix form. With $\mathbf{H}_F = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ and $\mathbf{H}_R = \mathbf{X}_R(\mathbf{X}_R^T\mathbf{X}_R)^{-1}\mathbf{X}_R^T$,

$$SS_{E(F)} = \mathbf{y}^T(\mathbf{I} - \mathbf{H}_F)\mathbf{y}, \qquad SS_{E(R)} = \mathbf{y}^T(\mathbf{I} - \mathbf{H}_R)\mathbf{y} \ \Rightarrow\ SS_H = \mathbf{y}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{y}.$$

Because $\Omega(\mathbf{X}_R) \subset \Omega(\mathbf{X})$, $\mathbf{H}_F\mathbf{X}_R = \mathbf{X}_R$, hence $\mathbf{H}_F\mathbf{H}_R = \mathbf{H}_R$ and (transposing) $\mathbf{H}_R\mathbf{H}_F = \mathbf{H}_R$. Then

$$(\mathbf{H}_F - \mathbf{H}_R)(\mathbf{H}_F - \mathbf{H}_R) = \mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_R + \mathbf{H}_R = \mathbf{H}_F - \mathbf{H}_R \quad \text{(idempotent)},$$
$$\operatorname{tr}(\mathbf{H}_F - \mathbf{H}_R) = (a + b - 1) - b = a - 1.$$

- $\mathrm{E}(SS_H) = \operatorname{tr}((\mathbf{H}_F - \mathbf{H}_R)\sigma^2\mathbf{I}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = \sigma^2(a-1) + b\sum_i \tau_i^2$, because $\boldsymbol{\xi} = (\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = (\mathbf{X} - \mathbf{H}_R\mathbf{X})\boldsymbol{\beta} = (\underbrace{\tau_1, \ldots, \tau_1}_{b}, \ldots, \underbrace{\tau_a, \ldots, \tau_a}_{b})^T$ (projecting on the block space removes μ and β_j and averages τ_i within a block, which gives 0). Hence $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + b\sum \tau_i^2/(a-1)$.
- $SS_{E(F)}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$ (Theorem 3.11 with $\mathbf{A} = (\mathbf{I} - \mathbf{H}_F)/\sigma^2$); $SS_H/\sigma^2 \sim \chi^2_{a-1}$ under H_0 ($\mathbf{B} = (\mathbf{H}_F - \mathbf{H}_R)/\sigma^2$).
- Independence (Theorem 3.12): $\mathbf{A}\sigma^2\mathbf{I}\mathbf{B} = \frac{1}{\sigma^2}(\mathbf{I} - \mathbf{H}_F)(\mathbf{H}_F - \mathbf{H}_R) = \frac{1}{\sigma^2}(\mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_F + \mathbf{H}_R) = \mathbf{0}$.

Therefore

$$F_0 = \frac{SS_H/(a-1)}{SS_{E(F)}/(a-1)(b-1)} = \frac{MS_{\text{Treatments}}}{MS_E} \overset{H_0}{\sim} F_{a-1,(a-1)(b-1)}.$$

Remark 4.6: R^2 never decreases when blocks are added (Exam 1, 2025)

Let $R^2 = 1 - SS_E/SS_T$. The full RCBD model (1) and the reduced model $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ (2) share the same SS_T. The reduced model's column space is contained in the full model's, so

$$SS_{E(1)} = \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \le \min_{\boldsymbol{\beta}_1} \|\mathbf{y} - \mathbf{X}_1\boldsymbol{\beta}_1\|^2 = SS_{E(2)}$$

(indeed $SS_{E(2)} - SS_{E(1)} = SS_{\text{Blocks}} \ge 0$ by the same argument as above). Hence $R^2_{(1)} \ge R^2_{(2)}$.

Remark 4.7: Omitting a real block effect biases the residuals (Exam 1, 2025)

Suppose the truth is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ (treatments + blocks) but we fit $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$. Then $\mathbf{e} = (\mathbf{I} - \mathbf{H}_1)\mathbf{y}$ and

$$\mathrm{E}(\mathbf{e}) = (\mathbf{I} - \mathbf{H}_1)(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) = (\mathbf{I} - \mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2.$$

$\mathbf{H}_1$ replaces each entry by its treatment average; the block effects average to $\frac{1}{b}\sum_j \beta_j = 0$ within every treatment, so $\mathbf{H}_1\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$ and $\mathrm{E}(e_{ij}) = \beta_j$. The block effects leak into the residuals and inflate SS_E (by $\mathrm{E}\,SS_{\text{Blocks}}$), destroying power.

Example 4.8: Hardness testing, line by line

$\sum\sum y_{ij}^2 = 1483.54$, $y_{\cdot\cdot}^2/N = 154^2/16 = 1482.25$.

$$\begin{aligned}
SS_T &= 1483.54 - 1482.25 = 1.29,\\
SS_{\text{Treatments}} &= \frac{38.3^2 + 38.4^2 + 37.8^2 + 39.5^2}{4} - 1482.25\\
&= \frac{5930.54}{4} - 1482.25 = 1482.635 - 1482.25 = 0.385,\\
SS_{\text{Blocks}} &= \frac{37.6^2 + 37.7^2 + 38.9^2 + 39.8^2}{4} - 1482.25 = \frac{5932.30}{4} - 1482.25 = 0.825,\\
SS_E &= 1.29 - 0.385 - 0.825 = 0.080.
\end{aligned}$$

Source	SS	DF	MS	F_0
Tips	0.385	3	0.1283	14.44
Coupons	0.825	3	0.2750	(30.94)
Error	0.080	9	0.00889	
Total	1.290	15		

$F_0 = 14.44 > F_{0.05,3,9} = 3.86$ ($P = 0.00087$): tips differ.
Consequence of not blocking: a one-way ANOVA puts SS_{Blocks} into error: $SS_E = 0.080 + 0.825 = 0.905$ on $9+3 = 12$ df, $MS_E = 0.0754$, $F_0 = 0.1283/0.0754 = 1.70$ ($P = 0.22$): the tip effect is no longer detected!

3.6 Residual analysis

Residuals $e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$: normal Q–Q plot; residuals vs treatment, vs block, vs fitted values. Levene's test can be applied between treatments ($z_{ij} = |y_{ij} - \tilde{y}_{i\cdot}|$) and between blocks ($z_{ij} = |y_{ij} - \tilde{y}_{\cdot j}|$); for the hardness data $P = 0.65$ and 0.68: no evidence of unequal variance.

4 Latin Square Design

4.1 Idea

- Controls (eliminates) **two** sources of nuisance variability simultaneously: rows and columns.
- Key assumption: the three factors (rows, columns, treatments) do **not interact**; otherwise the Latin square gives invalid results.
- A $p \times p$ Latin square: p treatments (Latin letters), two blocking variables with p levels each; each treatment appears *exactly once in each row and once in each column.*

Example with days (rows) and operators (columns):

	1	2	3
Mon	A	B	C
Tue	B	C	A
Wed	C	A	B

For $p = 3$ there are 12 distinct Latin squares (e.g. ABC/BCA/CAB, ACB/BAC/CBA, BAC/CBA/ACB, CBA/ACB/BAC, ...). In practice one chooses a square at random (randomize rows, columns and letter assignment).

4.2 Model

Definition 4.9

$$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}, \qquad i, j, k = 1, \ldots, p,$$

y_{ijk}: observation in row i, column k, treatment j; $\sum \alpha_i = \sum \tau_j = \sum \beta_k = 0$; $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Only p^2 of the p^3 index triples occur (the treatment index j is determined by (i, k)). The model is completely additive.

4.3 ANOVA decomposition

$$y_{ijk} - \bar{y}_{...} = (\bar{y}_{i..} - \bar{y}_{...}) + (\bar{y}_{..k} - \bar{y}_{...}) + (\bar{y}_{.j.} - \bar{y}_{...}) + (y_{ijk} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..k} + 2\bar{y}_{...}).$$

Squaring and summing over the p^2 cells, all product terms drop out:

$$SS_T = \underbrace{p \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2}_{SS_{\text{Rows}}} + \underbrace{p \sum_k (\bar{y}_{..k} - \bar{y}_{...})^2}_{SS_{\text{Columns}}} + \underbrace{p \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2}_{SS_{\text{Treatments}}} + SS_E,$$

$$p^2 - 1 = (p-1) + (p-1) + (p-1) + (p-2)(p-1).$$

Why the cross products vanish. Take e.g. rows $\times$ treatments: $\sum_{\text{cells}} (\bar{y}_{i..} - \bar{y})(\bar{y}_{.j.} - \bar{y}) = \sum_i (\bar{y}_{i..} - \bar{y}) \sum_{\text{cells in row } i} (\bar{y}_{.j.} - \bar{y})$. Row i contains every treatment exactly once, so the inner sum is $\sum_{j=1}^{p} (\bar{y}_{.j.} - \bar{y}) = 0$. The same "each level of one factor meets each level of another exactly once" (**orthogonality**) kills every main-effect cross product. For a main effect times the residual C_{ijk}, note that the residuals sum to zero over every row, every column and every treatment (e.g. in row i: $\sum_k (y_{ijk} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..k} + 2\bar{y}) = p\bar{y}_{i..} - p\bar{y}_{i..} - p\bar{y} - p\bar{y} + 2p\bar{y} = 0$). Error df: $p^2 - 1 - 3(p-1) = (p-1)(p+1-3) = (p-1)(p-2)$. ■

Source	SS	DF	MS	F_0
Treatments	$SS_{\text{Treatments}}$	$p-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Rows	SS_{Rows}	$p-1$	MS_{Rows}	
Columns	SS_{Columns}	$p-1$	MS_{Columns}	
Error	SS_E	$(p-2)(p-1)$	MS_E	
Total	SS_T	p^2-1		

Reject $H_0: \tau_1 = \cdots = \tau_p = 0$ if $F_0 > F_{\alpha,p-1,(p-2)(p-1)}$. As in the RCBD, $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + p\sum_j \tau_j^2/(p-1)$ and $\mathrm{E}(MS_E) = \sigma^2$ (same proof: $\bar{y}_{\cdot j\cdot} - \bar{y} = \tau_j + \bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}$ because each treatment meets every row and column once).

4.4 Least squares estimation

Derivation

$S(\theta) = \sum\sum\sum(y_{ijk} - \mu - \alpha_i - \beta_k - \tau_j)^2$. $\partial S/\partial\mu|_{\hat\theta} = -2\sum\sum\sum(y_{ijk} - \hat\mu - \hat\alpha_i - \hat\beta_k - \hat\tau_j) = 0 \Rightarrow p^2\hat\mu = y_{\cdots} \Rightarrow \hat\mu = \bar{y}_{\cdots}$. Similarly (using the zero-sum constraints and orthogonality)

$$\hat\alpha_i = \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}, \quad \hat\tau_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}, \quad \hat\beta_k = \bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}, \qquad \hat{y}_{ijk} = \bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdot\cdot k} - 2\bar{y}_{\cdots}.$$

Also $SS_T = SS_{\text{Model}} + SS_E$ with $SS_{\text{Model}} = SS_{\text{Rows}} + SS_{\text{Columns}} + SS_{\text{Treatments}}$.

E.g. for $\partial S/\partial\tau_j$: the p cells with treatment j lie in all p rows and all p columns, so $p\hat\mu + \sum_i \hat\alpha_i + p\hat\tau_j + \sum_k \hat\beta_k = y_{\cdot j\cdot} \Rightarrow \hat\tau_j = \bar{y}_{\cdot j\cdot} - \hat\mu$.

Example 4.10: Rocket propellant problem (5×5 Latin square)

Rows = batches of raw material, columns = operators, letters = formulations.

Batch	1	2	3	4	5	$y_{i\cdot\cdot}$
1	A=24	B=20	C=19	D=24	E=24	111
2	B=17	C=24	D=30	E=27	A=36	134
3	C=18	D=38	E=26	A=27	B=21	130
4	D=26	E=31	A=26	B=23	C=22	128
5	E=22	A=30	B=20	C=29	D=31	132
$y_{\cdot\cdot k}$	107	143	121	130	134	635

Treatment totals: $A = 143$, $B = 101$, $C = 112$, $D = 149$, $E = 130$.

$$\frac{y_{\cdots}^2}{N} = \frac{635^2}{25} = 16129, \qquad \sum y^2 = 16805, \qquad SS_T = 16805 - 16129 = 676,$$

$$SS_{\text{Batches}} = \frac{111^2+134^2+130^2+128^2+132^2}{5} - 16129 = \frac{80985}{5} - 16129 = 68,$$

$$SS_{\text{Operators}} = \frac{107^2+143^2+121^2+130^2+134^2}{5} - 16129 = \frac{81395}{5} - 16129 = 150,$$

$$SS_{\text{Formulations}} = \frac{143^2+101^2+112^2+149^2+130^2}{5} - 16129 = \frac{82295}{5} - 16129 = 330,$$

$$SS_E = 676 - 68 - 150 - 330 = 128.$$

Source	SS	DF	MS	F_0
Formulations	330	4	82.50	7.73
Batches	68	4	17.00	
Operators	150	4	37.50	
Error	128	12	10.67	
Total	676	24		

$F_0 = 82.5/10.67 = 7.73 > F_{0.05,4,12} = 3.26$ ($P = 0.0025$): formulations differ.
Fisher's LSD: $LSD = t_{0.025,12}\sqrt{2MS_E/p} = 2.179\sqrt{2(10.67)/5} = 2.179(2.066) = 4.50$.
Means: $\bar{y}_A = 28.6$, $\bar{y}_B = 20.2$, $\bar{y}_C = 22.4$, $\bar{y}_D = 29.8$, $\bar{y}_E = 26.0$.

Pair	|diff|	Sig.?	Pair	|diff|	Sig.?
A–B	8.4	*	B–D	9.6	*
A–C	6.2	*	B–E	5.8	*
A–D	1.2	–	C–D	7.4	*
A–E	2.6	–	C–E	3.6	–
B–C	2.2	–	D–E	3.8	–

Levene-type checks use $|y_{ijk} - \bar{y}_{i\cdot\cdot}|$ (rows), $|y_{ijk} - \bar{y}_{\cdot\cdot k}|$ (columns), $|y_{ijk} - \bar{y}_{\cdot j\cdot}|$ (treatments).

5 Graeco-Latin Square Design

- Controls **three** sources of nuisance variability: rows, columns and Greek letters.
- Superimpose two *orthogonal* Latin squares: one with Latin letters (treatments), one with Greek letters; each Greek letter appears exactly once in each row, once in each column, and exactly once with each Latin letter.
- Assumption: the four factors do not interact. (Graeco-Latin squares exist for every $p \geq 3$ except $p = 6$.)

Definition 4.11

$$y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}, \qquad i, j, k, l = 1, \ldots, p,$$

θ_i: row, τ_j: Latin treatment, ω_k: Greek letter, Ψ_l: column; all sum to zero.

$$y_{ijkl} - \bar{y} = (\bar{y}_{i\cdots} - \bar{y}) + (\bar{y}_{\cdot j\cdot\cdot} - \bar{y}) + (\bar{y}_{\cdot\cdot k\cdot} - \bar{y}) + (\bar{y}_{\cdots l} - \bar{y}) + (y_{ijkl} - \bar{y}_{i\cdots} - \bar{y}_{\cdot j\cdot\cdot} - \bar{y}_{\cdot\cdot k\cdot} - \bar{y}_{\cdots l} + 3\bar{y}),$$

$$SS_T = SS_{\text{Rows}} + SS_{\text{Columns}} + SS_{\text{Latin}} + SS_{\text{Greek}} + SS_E, \qquad p^2 - 1 = 4(p-1) + (p-3)(p-1).$$

Example 4.12: Rocket propellant with a Greek-letter factor (test assemblies)

Adding test assemblies (Greek letters) as a third blocking factor, with a standard orthogonal assignment

	1	2	3	4	5
1	$A\alpha$=24	$B\gamma$=20	$C\varepsilon$=19	$D\beta$=24	$E\delta$=24
2	$B\beta$=17	$C\delta$=24	$D\alpha$=30	$E\gamma$=27	$A\varepsilon$=36
3	$C\gamma$=18	$D\varepsilon$=38	$E\beta$=26	$A\delta$=27	$B\alpha$=21
4	$D\delta$=26	$E\alpha$=31	$A\gamma$=26	$B\varepsilon$=23	$C\beta$=22
5	$E\varepsilon$=22	$A\beta$=30	$B\delta$=20	$C\alpha$=29	$D\gamma$=31

Greek totals: $\alpha = 135$, $\beta = 119$, $\gamma = 122$, $\delta = 121$, $\varepsilon = 138$, so

$$SS_{\text{Greek}} = \frac{135^2 + 119^2 + 122^2 + 121^2 + 138^2}{5} - 16129 = \frac{80955}{5} - 16129 = 62,$$

and $SS_E = 676 - 68 - 150 - 330 - 62 = 66$ on $(5-3)(5-1) = 8$ df, $MS_E = 8.25$, $F_0 = 82.5/8.25 = 10.0 > F_{0.05,4,8} = 3.84$.
(An alternative R analysis, with $SS_{\text{Greek}} = 33.39$, $SS_E = 83.88$ on 8 df, $F_0 = 7.87$, was produced with a different Greek-letter coding in which batches/operators were no longer exactly orthogonal to the Greek factor — note its batch/operator SS changed from 68/150. In a genuine Graeco-Latin square all four SS are invariant to the order of fitting.)

6 Balanced Incomplete Block Design (BIBD)

6.1 Motivation and definitions

- Sometimes not all treatments fit in every block (e.g. each coupon can test only 3 of the 4 tips). An **incomplete block design** has more treatments than can be put in a single block.
- A **BIBD** is an incomplete block design in which *every pair of treatments occurs together the same number of times* (λ). (R. A. Fisher.)

Definition 4.13: BIBD parameters

a = # treatments, b = # blocks, k = # treatments per block, r = # replicates of each treatment, $N = ar = bk$, and

$$\lambda = \frac{r(k-1)}{a-1} = \#\text{ times each pair of treatments appears together.}$$

Counting arguments. (i) Count all observations two ways: by treatments, a treatments $\times$ r each $= ar$; by blocks, b blocks $\times$ k each $= bk$. So $N = ar = bk$. (ii) Fix one treatment, say 1. It appears in r blocks; in each such block it is paired with $k-1$ other treatments, giving $r(k-1)$ pairs containing treatment 1. Counted the other way, treatment 1 is paired with each of the other $a-1$ treatments exactly λ times: $\lambda(a-1)$. Hence $\lambda(a-1) = r(k-1)$. ■

A BIBD with a treatments and blocks of size k always exists by taking all $\binom{a}{k}$ combinations as blocks (unreduced design), although smaller designs often exist.

Example 4.14: Catalyst experiment

Catalyst	Batch (block)				
	1	2	3	4	$y_{i\cdot}$
1	73	74	–	71	218
2	–	75	67	72	214
3	73	75	68	–	216
4	75	–	72	75	222
$y_{\cdot j}$	221	224	207	218	$y_{\cdot\cdot} = 870$

$a = 4$, $b = 4$, $k = 3$, $r = 3$, $N = 12$, $\lambda = 3(2)/3 = 2$: e.g. catalysts 1 and 2 appear together in blocks 2 and 4.

6.2 Model and why an adjustment is needed

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2), \quad \sum \tau_i = \sum \beta_j = 0,$$

observed only when treatment i is in block j. Let $n_{ij} = 1$ if treatment i appears in block j and 0 otherwise. Then $\sum_j n_{ij} = r$, $\sum_i n_{ij} = k$, $\sum_j n_{ij} n_{lj} = \lambda$ $(i \neq l)$.

Derivation: the naive estimator is biased

RCBD ($a = b = 4$, nothing missing): $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ and

$$\mathrm{E}(\hat{\tau}_i) = \tfrac{1}{4}\mathrm{E}(y_{i1}+y_{i2}+y_{i3}+y_{i4}) - \tfrac{1}{16}\mathrm{E}(y_{11}+\cdots+y_{44}) = \tfrac{1}{4}(4\mu+4\tau_i) - \tfrac{1}{16}(16\mu+\textstyle\sum\tau_i+\sum\beta_j) = \tau_i.$$

BIBD (catalyst layout, treatment 1 in blocks 1, 2, 4):

$$\begin{aligned} \mathrm{E}(y_{1\cdot}) &= \mathrm{E}(y_{11}+y_{12}+y_{14}) = 3\mu + 3\tau_1 + (\beta_1+\beta_2+\beta_4), \\ \mathrm{E}(y_{\cdot\cdot}) &= N\mu + r\textstyle\sum_i \tau_i + k\sum_j \beta_j = N\mu, \\ \mathrm{E}(\bar{y}_{1\cdot} - \bar{y}_{\cdot\cdot}) &= \mu + \tau_1 + \tfrac{1}{3}(\beta_1+\beta_2+\beta_4) - \mu = \tau_1 + \tfrac{1}{3}(\beta_1+\beta_2+\beta_4) \neq \tau_1. \end{aligned}$$

Because each treatment sees a different set of r blocks, treatment totals are contaminated by block effects: the treatment SS must be **adjusted for blocks**.

6.3 Least squares: normal equations

Derivation

$S(\theta) = \sum_i \sum_j n_{ij}(y_{ij} - \mu - \tau_i - \beta_j)^2$.

$$\begin{aligned} \frac{\partial S}{\partial \mu} = 0: &\quad N\hat{\mu} + r\sum_{i=1}^{a}\hat{\tau}_i + k\sum_{j=1}^{b}\hat{\beta}_j = y_{\cdot\cdot} \quad \Rightarrow \quad \hat{\mu} = y_{\cdot\cdot}/N = \bar{y}_{\cdot\cdot}, \\ \frac{\partial S}{\partial \tau_i} = 0: &\quad r\hat{\mu} + r\hat{\tau}_i + \sum_{j=1}^{b} n_{ij}\hat{\beta}_j = y_{i\cdot}, \qquad i = 1, \ldots, a, \\ \frac{\partial S}{\partial \beta_j} = 0: &\quad k\hat{\mu} + \sum_{i=1}^{a} n_{ij}\hat{\tau}_i + k\hat{\beta}_j = y_{\cdot j}, \qquad j = 1, \ldots, b \quad (\star). \end{aligned}$$

These are $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$.

Theorem 4.15: Adjusted treatment effects

$$\hat{\tau}_i = \frac{k\,Q_i}{\lambda a}, \qquad Q_i = y_{i\cdot} - \frac{1}{k}\sum_{j=1}^{b} n_{ij} y_{\cdot j} \quad \text{(adjusted treatment total)}, \qquad \sum_{i=1}^{a} Q_i = 0.$$

Proof. Step 1 (eliminate $\hat{\beta}$). Multiply the τ_i-equation by k:

$$rk\hat{\mu} + rk\hat{\tau}_i + k\sum_{j} n_{ij}\hat{\beta}_j = k y_{i\cdot}. \tag{a}$$

Multiply $(\star)$ by n_{ij} and sum over j:

$$k\hat{\mu}\sum_{j} n_{ij} + \sum_{j}\sum_{l} n_{ij} n_{lj}\hat{\tau}_l + k\sum_{j} n_{ij}\hat{\beta}_j = \sum_{j} n_{ij} y_{\cdot j}. \tag{b}$$

Since $\sum_j n_{ij} = r$, the first term of (b) is $rk\hat{\mu}$. Subtract (b) from (a): the $\hat{\mu}$ and $\hat{\beta}$ terms cancel,

$$rk\hat{\tau}_i - \sum_{j=1}^{b}\sum_{l=1}^{a} n_{ij}n_{lj}\hat{\tau}_l = ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j} = kQ_i.$$

Step 2 (use balance). Split the double sum into $l = i$ and $l \neq i$, using $n_{ij}^2 = n_{ij}$:

$$\sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l = \underbrace{\sum_j n_{ij}}_{=r}\hat{\tau}_i + \sum_{l\neq i}\underbrace{\Big(\sum_j n_{ij}n_{lj}\Big)}_{=\lambda}\hat{\tau}_l = r\hat{\tau}_i + \lambda\sum_{l\neq i}\hat{\tau}_l = r\hat{\tau}_i - \lambda\hat{\tau}_i,$$

using $\sum_l \hat{\tau}_l = 0 \Rightarrow \sum_{l\neq i}\hat{\tau}_l = -\hat{\tau}_i$. **Step 3.** Therefore

$$rk\hat{\tau}_i - r\hat{\tau}_i + \lambda\hat{\tau}_i = kQ_i \iff (r(k-1)+\lambda)\hat{\tau}_i = kQ_i \iff (\lambda(a-1)+\lambda)\hat{\tau}_i = kQ_i \iff \hat{\tau}_i = \frac{kQ_i}{\lambda a}.$$

$\sum Q_i = 0$**:** $\sum_i Q_i = \sum_i y_{i\cdot} - \frac{1}{k}\sum_j (\sum_i n_{ij})y_{\cdot j} = y_{\cdot\cdot} - \frac{1}{k}\cdot k\, y_{\cdot\cdot} = 0.$ ■

Theorem 4.16: Unbiasedness, $\mathrm{E}(\hat{\tau}_i) = \tau_i$ (Final 2025)

Proof. $\mathrm{E}(y_{i\cdot}) = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$ and $\mathrm{E}(y_{\cdot j}) = k\mu + \sum_l n_{lj}\tau_l + k\beta_j$. Then

$$\frac{1}{k}\sum_j n_{ij}\mathrm{E}(y_{\cdot j}) = \frac{1}{k}\Big[kr\mu + \sum_j\sum_l n_{ij}n_{lj}\tau_l + k\sum_j n_{ij}\beta_j\Big] = r\mu + \frac{1}{k}\big[r\tau_i + \lambda\textstyle\sum_{l\neq i}\tau_l\big] + \sum_j n_{ij}\beta_j$$
$$= r\mu + \frac{(r-\lambda)\tau_i}{k} + \sum_j n_{ij}\beta_j,$$

so

$$\mathrm{E}(Q_i) = r\tau_i - \frac{(r-\lambda)\tau_i}{k} = \frac{r(k-1)+\lambda}{k}\tau_i = \frac{\lambda a}{k}\tau_i \quad\Rightarrow\quad \mathrm{E}(\hat{\tau}_i) = \frac{k}{\lambda a}\cdot\frac{\lambda a}{k}\tau_i = \tau_i.$$

The block effects $\sum_j n_{ij}\beta_j$ cancel exactly — that is the meaning of "adjusted". ■

6.4 Sums of squares

Theorem 4.17

$$SS_T = \sum_i\sum_j y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Blocks}} = \frac{1}{k}\sum_{j=1}^{b} y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Treatments(adjusted)}} = \frac{k\sum_{i=1}^{a} Q_i^2}{\lambda a},$$
$$SS_E = SS_T - SS_{\text{Treatments(adjusted)}} - SS_{\text{Blocks}}.$$

SS_{Blocks}**.** Every block has k observations, so

$$SS_{\text{Blocks}} = k\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 = k\Big[\sum_j \bar{y}_{\cdot j}^2 - b\bar{y}_{\cdot\cdot}^2\Big] = k\Big[\sum_j \frac{y_{\cdot j}^2}{k^2} - b\frac{y_{\cdot\cdot}^2}{b^2k^2}\Big] = \frac{1}{k}\sum_j y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{bk},$$

and $bk = N$. (Blocks are *unadjusted*: fitted first.) ■

$SS_{\text{Treatments(adjusted)}}$**.** **Model SS.** $SS_T = SS_{\text{Model}} + SS_E$ with $SS_T = \mathbf{y}^T\mathbf{y} - \frac{1}{N}y_{\cdot\cdot}^2$ and

$$SS_E = (\mathbf{y}-\hat{\mathbf{y}})^T(\mathbf{y}-\hat{\mathbf{y}}) = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\hat{\mathbf{y}} - \hat{\mathbf{y}}^T\mathbf{y} + \hat{\mathbf{y}}^T\hat{\mathbf{y}} = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$$

(because $\hat{\mathbf{y}}^T\hat{\mathbf{y}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$ by the normal equations). Hence

$$SS_{\text{Model}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{1}{N}y_{\cdot\cdot}^2 = \hat{\mu}y_{\cdot\cdot} + \sum_{i=1}^{a} y_{i\cdot}\hat{\tau}_i + \sum_{j=1}^{b} y_{\cdot j}\hat{\beta}_j - \frac{1}{N}y_{\cdot\cdot}^2.$$

Eliminate $\hat{\beta}_j$. Multiply $(\star)$ by $y_{\cdot j}$: $k\hat{\mu}y_{\cdot j} + \sum_i n_{ij}\hat{\tau}_i y_{\cdot j} + k\hat{\beta}_j y_{\cdot j} = y_{\cdot j}^2$, so

$$\sum_j \hat{\beta}_j y_{\cdot j} = \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}\sum_j y_{\cdot j} - \frac{1}{k}\sum_j\sum_i n_{ij}\hat{\tau}_i y_{\cdot j}.$$

Subtract blocks. With $\hat{\mu}\sum_j y_{\cdot j} = \hat{\mu}y_{\cdot\cdot}$,

$$\begin{aligned} SS_{\text{Model}} - SS_{\text{Blocks}} &= \hat{\mu}y_{\cdot\cdot} + \sum_i y_{i\cdot}\hat{\tau}_i + \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}y_{\cdot\cdot} - \frac{1}{k}\sum_j\sum_i n_{ij}\hat{\tau}_i y_{\cdot j} - \frac{1}{N}y_{\cdot\cdot}^2 - \frac{1}{k}\sum_j y_{\cdot j}^2 + \frac{1}{N}y_{\cdot\cdot}^2 \\ &= \sum_{i=1}^{a}\Big(y_{i\cdot} - \frac{1}{k}\sum_j n_{ij}y_{\cdot j}\Big)\hat{\tau}_i = \sum_i Q_i\hat{\tau}_i = \sum_i Q_i\cdot\frac{k}{\lambda a}Q_i = \frac{k}{\lambda a}\sum_{i=1}^{a} Q_i^2. \end{aligned}$$

So $SS_{\text{Treatments(adjusted)}} = SS_{\text{Model}} - SS_{\text{Blocks}}$ is the extra SS due to treatments after blocks. ■

Source	SS	DF	MS	F_0
Treatments (adjusted)	$\frac{k\sum Q_i^2}{\lambda a}$	$a-1$	$MS_{\text{Trt(adj)}}$	$MS_{\text{Trt(adj)}}/MS_E$
Blocks	$\frac{1}{k}\sum y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}$	$b-1$	MS_{Blocks}	
Error	by subtraction	$N-a-b+1$	MS_E	
Total	$\sum\sum y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}$	$N-1$		

Reject $H_0: \tau_i = 0$ if $F_0 > F_{\alpha,a-1,N-a-b+1}$.

6.5 Variances of the adjusted estimators

Theorem 4.18 (Midterm 2, 2025)

$$\text{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2, \qquad \text{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2 \ (i \neq l),$$
$$\text{Var}(\hat{\tau}_i) = \frac{k(a-1)}{\lambda a^2}\sigma^2, \qquad \text{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{2k}{\lambda a}\sigma^2.$$

Proof. Splitting argument for $\text{Var}(Q_i)$. Write

$$kQ_i = ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j} = (k-1)\underbrace{y_{i\cdot}}_{\text{obs. of trt } i} - \underbrace{\Big(\sum_j n_{ij}y_{\cdot j} - y_{i\cdot}\Big)}_{\text{other obs. in those } r \text{ blocks}}.$$

The two pieces involve disjoint sets of observations, hence are independent. $y_{i\cdot}$ is a sum of r observations; the second piece is a sum of $r(k-1)$ observations. So

$$k^2\text{Var}(Q_i) = (k-1)^2 r\sigma^2 + r(k-1)\sigma^2 = r(k-1)k\sigma^2 \ \Rightarrow\ \text{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2.$$

Covariance. For $i \neq l$, using $\text{Cov}(y_{i\cdot}, y_{\cdot j}) = n_{ij}\sigma^2$, $\text{Var}(y_{\cdot j}) = k\sigma^2$, $\text{Cov}(y_{i\cdot}, y_{l\cdot}) = 0$, $\text{Cov}(y_{\cdot j}, y_{\cdot j'}) = 0$ $(j \neq j')$:

$$\text{Cov}(kQ_i, kQ_l) = \text{Cov}\Big(ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j},\ ky_{l\cdot} - \sum_j n_{lj}y_{\cdot j}\Big)$$

$$= 0 - k\sum_j n_{lj}\mathrm{Cov}(y_{i\cdot}, y_{\cdot j}) - k\sum_j n_{ij}\mathrm{Cov}(y_{\cdot j}, y_{l\cdot}) + \sum_j n_{ij}n_{lj}\mathrm{Var}(y_{\cdot j})$$

$$= -k\lambda\sigma^2 - k\lambda\sigma^2 + \lambda k\sigma^2 = -k\lambda\sigma^2 \;\Rightarrow\; \mathrm{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2.$$

Variances of $\hat{\tau}$. With $r(k-1) = \lambda(a-1)$:

$$\mathrm{Var}(\hat{\tau}_i) = \frac{k^2}{\lambda^2 a^2}\mathrm{Var}(Q_i) = \frac{k^2}{\lambda^2 a^2}\cdot\frac{\lambda(a-1)}{k}\sigma^2 = \frac{k(a-1)}{\lambda a^2}\sigma^2,$$

$$\mathrm{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{k^2}{\lambda^2 a^2}[\mathrm{Var}Q_i + \mathrm{Var}Q_l - 2\mathrm{Cov}(Q_i, Q_l)] = \frac{k^2}{\lambda^2 a^2}\Big[\frac{2\lambda(a-1)}{k} + \frac{2\lambda}{k}\Big]\sigma^2 = \frac{2k}{\lambda a}\sigma^2.$$

■

Remark 4.19

Compare with an RCBD with r replicates: $\mathrm{Var}(\hat{\tau}_i - \hat{\tau}_l) = 2\sigma^2/r$. The ratio $\frac{2\sigma^2/r}{2k\sigma^2/(\lambda a)} = \frac{\lambda a}{rk}$ is the **efficiency factor** of the BIBD (< 1). Multiple comparisons on adjusted means use the standard error $\sqrt{2kMS_E/(\lambda a)}$.

Example 4.20: Catalyst BIBD, line by line

$\sum\sum y_{ij}^2 = 63156$, $y_{\cdot\cdot}^2/N = 870^2/12 = 63075$.

$$SS_T = 63156 - 63075 = 81.00,$$

$$SS_{\text{Blocks}} = \frac{221^2 + 224^2 + 207^2 + 218^2}{3} - 63075 = \frac{189390}{3} - 63075 = 55.00.$$

Adjusted totals (block totals of the blocks containing treatment i):

$$Q_1 = 218 - \tfrac{1}{3}(221 + 224 + 218) = 218 - \tfrac{663}{3} = -3.00,$$
$$Q_2 = 214 - \tfrac{1}{3}(224 + 207 + 218) = 214 - \tfrac{649}{3} = -2.33,$$
$$Q_3 = 216 - \tfrac{1}{3}(221 + 224 + 207) = 216 - \tfrac{652}{3} = -1.33,$$
$$Q_4 = 222 - \tfrac{1}{3}(221 + 207 + 218) = 222 - \tfrac{646}{3} = 6.67 \qquad (\textstyle\sum Q_i = 0\ \checkmark),$$

$$SS_{\text{Trt(adj)}} = \frac{3\,[(-3)^2 + (-7/3)^2 + (-4/3)^2 + (20/3)^2]}{2\cdot 4}$$
$$= \frac{3(9 + 5.444 + 1.778 + 44.444)}{8} = \frac{3(60.667)}{8} = 22.75,$$

$$SS_E = 81.00 - 22.75 - 55.00 = 3.25.$$

Source	SS	DF	MS	F_0
Treatments (adj.)	22.75	3	7.58	11.67
Blocks	55.00	3	18.33	
Error	3.25	5	0.65	
Total	81.00	11		

$F_0 = 7.58/0.65 = 11.67 > F_{0.05,3,5} = 5.41$ ($P = 0.0107$): catalysts differ. $\hat{\tau}_i = kQ_i/(\lambda a) = 3Q_i/8$: $\hat{\tau} = (-1.125, -0.875, -0.500, 2.500)$. $\widehat{\mathrm{se}}(\hat{\tau}_i - \hat{\tau}_l) = \sqrt{2(3)(0.65)/(2\cdot 4)} = 0.698$.

R and the order of terms. `lm(y~block+trt)` gives sequential SS: blocks (unadjusted) 55.00, treatments adjusted for blocks 22.75 — the correct BIBD analysis. `lm(y~trt+block)` gives treatments *unadjusted* $= \sum y_{i\cdot}^2/r - y_{\cdot\cdot}^2/N = 11.667$ and blocks adjusted $= 66.083$; the error SS (3.25) is the same. In incomplete designs the order matters because treatments and blocks are not orthogonal.

Example 4.21: Constructing a BIBD (Midterm 2, 2025)

$a = 5$ gasoline additives, $b = 5$ cars (blocks), $k = 4$ additives per car.

$$N = bk = 5 \cdot 4 = 20, \qquad r = \frac{N}{a} = \frac{20}{5} = 4, \qquad \lambda = \frac{r(k-1)}{a-1} = \frac{4 \cdot 3}{4} = 3.$$

Design: block j omits additive j:

$$\text{B1} : \{2,3,4,5\},\ \text{B2} : \{1,3,4,5\},\ \text{B3} : \{1,2,4,5\},\ \text{B4} : \{1,2,3,5\},\ \text{B5} : \{1,2,3,4\}.$$

Check: the pair $(1,2)$ appears in every block except B1 and B2, i.e. in $5-2=3=\lambda$ blocks; the same holds for every pair.

7 Summary

Design	Nuisance factors	Error df
RCBD	1 (blocks)	$(a-1)(b-1)$
Latin square	2 (rows, columns)	$(p-1)(p-2)$
Graeco-Latin square	3	$(p-1)(p-3)$
BIBD	1 (incomplete blocks)	$N-a-b+1$